

The market for corporate control

Week 41 — why the one bidder on earth still earns nothing

Advanced Corporate Finance (EBM232A05)

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Where we are

Four rows. Every friction so far lived *inside* one firm:

Wk	Friction	Effect	Evidence
36	Personal taxes (Miller)	value $\uparrow / \downarrow$	net benefit (Korteweg, 2010)
37	Dynamics / flexibility	leverage $\downarrow$	flexibility survey (Graham and Harvey, 2001)
38	Costly external finance	investment $\downarrow$	cash hoarding (Almeida et al., 2004)
39	The manager wedge	risk-taking $\downarrow$	weak incentives (Jensen and Murphy, 1990)
41	Free-riding on control	? today	—

Last time we left a manager doing the wrong thing and nobody able to stop him. The obvious remedy is that **somebody buys the firm**. Today: why that remedy is **not** self-financing.

- The half you already computed: synergy, the payment method, and why **competition** leaves the acquirer at zero.
- The puzzle: the competition story **fails its own test** — almost no deal has a second bidder.
- The twist: a raider who is the **only** bidder on earth also earns exactly zero — because the *target's* shareholders free-ride.
- The fix: **appropriability** — a toehold, or a charter right to dilute — and how far it overshoots.
- Evidence: what large acquirer losses look like, and one test that separates the two stories.

One friction — **the target has many owners** — and a familiar fact acquires the opposite welfare sign.

The half you already computed

You have priced a takeover

From the bachelor, and you computed all of it: the deal is worth doing iff the combination beats the parts, $V_{AT} > V_A + V_T$; a cash offer and a share offer are **not** the same instrument; and **EPS accretion is not value** — a share-financed deal can raise earnings per share while creating nothing.

- You also know the punchline of the evidence: **targets gain a lot, acquirers roughly nothing**.
- And you were given a reason: **competition**. Bidders bid against each other until the surplus has been handed to the target.

None of that is re-taught today. Today we ask whether the **reason** is right — and what follows if it is not.

Why the competition story is comforting

If competition among bidders is what drives the acquirer's return to zero, then the takeover market is working exactly as it should:

- The deal still **happens**. The value improvement is realised.
- The surplus is merely **redistributed** — from the acquirer's owners to the target's.
- Zero acquirer returns are then a **distributional** fact, not an efficiency problem. Nothing is lost.

Hold on to that last line. Everything today turns on whether a zero return means the surplus was **transferred** — or **never created**.

**The puzzle: the disciplinarian earns
nothing**

Two facts you already believe

Around the announcement	Return	Source
Target, window $(-1, +1)$	+16%	(Andrade et al., 2001), 1973–98
Target, -20 days to completion	+23.8%	(Andrade et al., 2001)
Acquirer	-0.7%	(Andrade et al., 2001)
Acquirer, deals after 2009	+1.05%	(Alexandridis et al., 2017)

Twenty-five years of data, one shape: the target captures the gains and the acquirer is left at, or just below, break-even.

The fact is not in dispute. The **mechanism** is, and the mechanism is what determines whether anything was lost.

The explanation fails its own test

If **rival bidders** bid the surplus away, then deals *without* a rival should leave the acquirer with a visible gain. So how many deals have a rival?

Share of acquisitions facing a competing bid, 1980–97	1.4%
... among the very largest value-destroying deals	8.0%

Moeller et al. (2005), Table 3.

Ninety-nine deals in a hundred are uncontested — and the acquirer still earns nothing.

So competition cannot be carrying this fact. Something else drives the acquirer's return to zero, and it is not another bidder.

The free-rider

One new ingredient: the target has many owners

Every model so far had **one** decision-maker per side. A public target does not: it has thousands of small holders, and **each one decides separately** whether to sell.

- A raider can lift the firm's value per share from v to $v' > v$ — better management, a closed division, an ended empire. Doing so costs him c .
- He announces a tender offer at price p , conditional on getting control.
- Every holder now answers one question: **do I sell at p , or keep my share?**

Note what this friction is **not**. Nobody is uninformed and nobody is irrational. Every holder knows exactly what the raider will do.

What one small holder actually decides

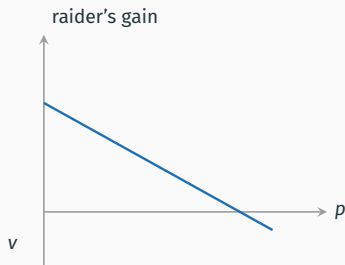
He is **not pivotal**. The offer is conditional on control, and one share in millions does not decide whether it succeeds — so he takes the outcome as given:

	He tenders	He refuses
Offer succeeds	p	v'
Offer fails	v	v

Expecting success, he tenders only if $p \geq v'$. Below that, refusing dominates: he keeps a share the raider just made **more valuable**, and pays none of c .

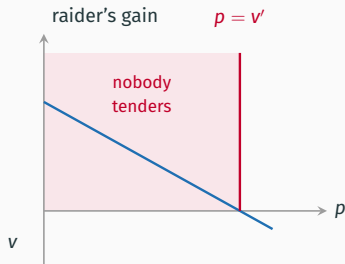
Would you tender at €14 knowing the share is worth €15 tomorrow? If not — you have just derived the result.

What the raider can capture



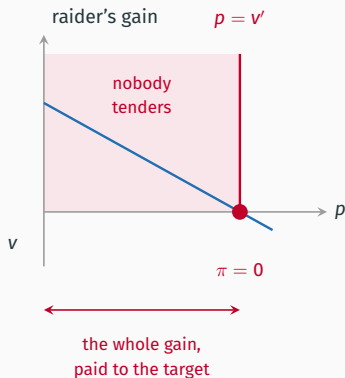
Start where a bachelor NPV starts. He offers p , he ends up owning shares worth v' , so on every share he buys he keeps $v' - p$. Offer just above the current price v and that margin is nearly the whole improvement.

What the raider can capture



Now impose what the holder just told us. At **any** price below v' , refusing dominates — so in the entire shaded region **not a single share is offered to him**. Every price at which he would profit is a price at which he buys nothing.

What the raider can capture



One price survives, and at that price his margin is exactly **zero**. He must hand the entire improvement $v' - v$ across to the people he is buying from — and he is still out of pocket by his own cost c . So he does not bid.

The market for control is a public good

Tendering requires $p \geq v'$, so the raider's profit is

$$\pi = (v' - p) - c \leq -c < 0 \quad \implies \quad \text{no bid.}$$

He produces the improvement; **every** shareholder consumes it; **none** of them pays for it. That is a public good, and public goods are **underprovided**.

- Note there is **no second bidder anywhere in this argument**. A monopolist raider earns zero too.
- And the welfare sign is the opposite of the bachelor's: the gain is not redistributed, it is **never created**.

Same observation — acquirers earn nothing — and now it means good takeovers **are not happening** (Grossman and Hart, 1980).

Put in numbers

A firm worth €10 a share. A raider who would run it better: $v' = €15$. His cost of doing it: $c = €0.95$ a share.

		Per share
Improvement he creates	$v' - v$	+5.00
Lowest price at which anyone sells	$p = v'$	15.00
What he keeps on each share	$v' - p$	0.00
His profit	$-c$	-0.95
Value that is therefore never created		5.00

Fifty percent of the firm's value, sitting on the table, in a market with no taxes, no distress costs and no asymmetric information. **Nobody picks it up.**

The fix, and its overshoot

Fix 1 — own some of it before you announce

Buy a stake α — a **toehold** — quietly, at the old price v , and only then bid. The shares he already holds appreciate by $v' - v$, and nobody free-rides on those.

$$\pi = \alpha(v' - v) - c \quad \implies \quad \text{he bids iff } \alpha \geq \frac{c}{v' - v}.$$

In our numbers: $\alpha \geq 0.95/5 = 19\%$ of the company, bought before anyone notices.

The economics has not changed — he still cannot take a cent from the shares he buys *in* the offer. He can only profit on what he owned **beforehand**.

And the regulator sets the ceiling

Nineteen percent, quietly, is not something securities law permits.

	Disclose at	Deadline
United States (Williams Act, §13(d))	5%	5 business days
Netherlands (Wft, ch. 5.3)	3%	notify the AFM

Once he files, the price jumps to reflect the improvement, and every share bought afterwards earns him nothing. So toeholds should be **rare**, and **large** where they occur — which is what Betton et al. (2009) find, with an estimated threshold averaging **9%** across 10,000 control bids.

A rule written to protect shareholders also decides **which improvements are worth making**.

Fix 2 — the right to dilute

Grossman and Hart's own answer. Let the charter permit the winner to extract φ per share from whoever *did not* tender — a freeze-out merger, a squeeze-out, a transfer on non-market terms.

- Refusing is now worth $v' - \varphi$, not v' . So holders tender at $p = v' - \varphi$, and the raider keeps φ on every share he buys.

$$\pi = \underbrace{\alpha (v' - v)}_{\text{his toehold}} + \underbrace{(1 - \alpha) \varphi}_{\text{what he may take}} - c.$$

Read that carefully, because it says something strange, and the next slide is about why it is nevertheless right.

Why shareholders would write that rule

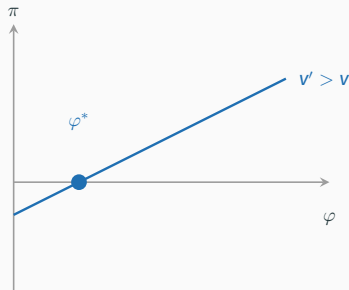
The charter is chosen **before** any raider appears — and before you know whether you would be the holder who tenders or the one who refuses.

Charter	Ex post	Ex ante
$\varphi = 0$	your shares are safe	no bid ever comes
$\varphi > 0$	he takes φ from you	a credible threat

The expropriation is real, and shareholders write it in anyway — because the alternative is a manager who knows he can never be removed.

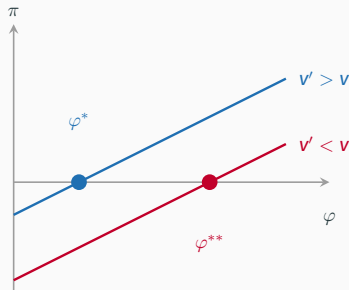
Ex post it is a loss; ex ante it is the price of discipline. Squeeze-out rules put the same provision in the law rather than the charter.

How much dilution should the charter allow?



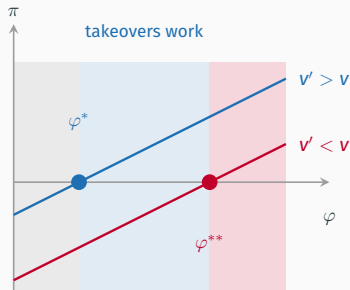
Take the raider we want, the one with $v' > v$. His profit rises one-for-one with the right to dilute that the charter hands him. Below φ^* he does not bid at all — so a charter that protects minorities perfectly buys **no discipline whatsoever**.

How much dilution should the charter allow?



Now put a raider who would **destroy** value on the same dial. Same slope — the dilution right pays him just as well — only a lower intercept, because his toehold loses money. He crosses zero too, later, at ϕ^{**} . What separates the two lines is **only the toehold** — α times the gap between the two raiders' values. At $\alpha = 0$ they **coincide**.

How much dilution should the charter allow?



Three regions, and the charter must be written before anyone knows which raider will show up. Too little: nothing happens. Too much: the takeover market runs on **plunder**. Only the middle band buys discipline without buying looting.

... and the overshoot is not hypothetical

A 10% toehold; our raider adds €5 a share, a rival **destroys** €4:

Dilution the charter allows	$v' = €15$	$v' = €6$
$\varphi = €0.25$	does not bid	does not bid
$\varphi = €1.00$	bids	does not bid
$\varphi = €2.00$	bids	bids
Threshold	$\varphi^* = €0.50$	$\varphi^{**} = €1.50$

The instrument is **blind**: it pays for control, not for improvement.

Week 39 again — the cure for one distortion is an instrument that manufactures the opposite one. Poison pills dial φ back *down*, so whether a defence entrenches or protects depends on which band you were in. **The taxonomy is Hermes's course**; the dial is ours.

Evidence

When acquirers lose, they lose big

Moeller et al. (2005) take 4,136 US acquisitions and ask where the aggregate losses sit.

“Large-loss” deals (acquirer loses $>$ \$1bn on announcement)	87
Their combined loss	\$397bn
Loss per dollar spent	\$2.31
Aggregate gain of the <i>other</i> 4,049 deals	+\$157bn

Those 87 are large, public-target, **equity-financed** deals at the top of the 1998–2001 market — and they are not explained by hostility, diversification, or size alone.

“Acquirers break even on average” is an average over a distribution with a very long left tail. Two percent of deals carry the entire aggregate loss.

The test that separates the two stories

Both stories predict an acquirer return of about zero, and disagree about everything else:

	Competition	Free-riding
Needs a rival bidder	yes	no
Share of deals with a rival	should be high	1.4% observed
Uncontested deal	acquirer gains	acquirer still at zero
The value improvement	realised	often never attempted

The deals you observe are not the population — they are the ones that cleared the appropriability hurdle.

So announcement returns cannot settle whether takeovers create value.

The other reason deals happen

Waves, and the currency

Free-riding explains why disciplinary deals are **scarce**. It does not explain why the deals we *do* see arrive in clusters, at market peaks, paid for in **stock**.

- Shleifer and Vishny (2003) invert the usual reading: the acquirer's overvalued equity is not a way of *paying* — it is the **reason for the deal**. A manager with a temporarily overpriced currency converts it into real assets.
- Then a value-*destroying* merger can still be rational for the acquirer's long-horizon shareholders, because the alternative was holding an asset the market would eventually reprice.
- Consistent with Moeller et al. (2005)'s 87: large, equity-financed, at the top of a market.

Note the register change. Grossman–Hart is a friction in **ownership**; this is a friction in **prices**. Both produce mergers; only one of them is about control.

Which story is it?

The two accounts make opposing predictions, and Harford (2005) runs them against each other:

Prediction	Misvaluation	Industry shock
Pre-wave returns	high	high
Dispersion of those returns	high	low
Cash-paid, partial-firm deals	absent	present
Capital-market liquidity	irrelevant	must be loose

His verdict leans to the shock story *conditional on* cheap capital — misvaluation gets the timing, but not the wave.

The habit is the point, and the project asks it of you: when two theories predict the same headline number, find the margin on which they **disagree**.

Ledger & wrap

A zero return to acquirers is not evidence that the takeover market works. It is exactly what you would see if the takeover market **could not pay for itself** — and then the deals you are measuring are the survivors, not the sample.

Two things to carry forward:

- “Identify the friction” now includes **how many people must agree**. Dispersed ownership is itself a friction, and it defeats a mechanism that a single owner would have run happily.
- Every fix for it works by making the improvement **appropriable** — and an instrument that pays for control cannot tell a good raider from a bad one.

The frictions ledger — Week 41

Five rows. Bring this back next week.

Wk	Friction	Effect	Evidence
36	Personal taxes (Miller)	value $\uparrow$ / $\downarrow$	net benefit (Korteweg, 2010)
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41	Free-riding on control	takeovers $\downarrow$	acquirer gains ≈ 0 (Moeller et al., 2005)

This is the first row whose effect is a **non-event**: the friction shows up in the deals that did not happen. Which is exactly why it took a model to see it.

What is new here — and what is not

You already own:

- synergy, cash vs. stock, the EPS fallacy
- “acquirers break even” as a fact
- overconfident managers overpaying

This week adds:

- the **free-rider**, and a zero that means *underprovision*
- **appropriability** — toehold, dilution, and their ceiling
- takeover defences as a **dial**, not a taxonomy

The fact you arrived with is unchanged. What changed is what it is evidence **of** — and that is the whole difference between a market that redistributes and a market that fails.

Reading, and next

- **Reading:** Grossman and Hart (1980) — read Sections 1–2 for the argument; the algebra above is its two-state version. Then Moeller et al. (2005) for the evidence.
- **If you want the wave story:** Shleifer and Vishny (2003), with Harford (2005) as the referee.
- **This week:** the **project midpoint check-in** — one slot per team; bring what runs, not what you meant to write. The tutorial covers **event-study mechanics**: how every number in this lecture was produced.
- **Next:** Week 42. If the market cannot discipline a firm by itself, somebody writes rules instead — hedging, deposit insurance, and capital as a deductible.

Everybody wants the firm fixed; nobody wants to be the one who sells.

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